

# A Modulus-Complete Spectral-Tail Certificate for the Noisy Hardy Determinant

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## Abstract

The fixed-rank natural- $L^2$  quotient fails at zero noise, while the variable-rank criterion of RH-279 was previously uninstantiated. We activate its tail branch without constructing a Riesz projector. After removing the two peripheral eigenvalues of the Hardy-scaled folded Gaussian operator, take every algebraic eigenvalue of modulus greater than  $q = 1/2$  as the spectral head and put the remaining eigenvalues on a normal diagonal operator  $C_\sigma$ . The sharp Hilbert–Schmidt mass law bounds their squared sum by  $\sigma^{-1}$  for small noise. With  $R = 7/5$  and  $m_\sigma = \lceil 4\log(1/\sigma) \rceil$ , the trace norm, operator norm, prefix, and root-rate hypotheses of RH-279 hold, and the full logarithmic tail beyond  $m_\sigma$  vanishes uniformly on  $|z| \leq R$ . This proves a projection-free spectral-tail theorem. It does not identify the noisy head with the monodromy counterloop.

## 1 Spectral realization

Let  $A_\sigma = K_\sigma/r_H$ ,  $r_H = 0.85$ , and remove the Perron and negative-parity eigenvalues from its algebraic spectrum. Denote the remaining sequence, counted with algebraic multiplicity, by  $(\mu_j(\sigma))_{j \geq 1}$ . RH-276 and the Hilbert–Schmidt eigenvalue inequality give, for sufficiently small  $\sigma$ ,

$$M_\sigma := \sum_j |\mu_j(\sigma)|^2 \leq \|A_\sigma\|_{\mathcal{S}_2}^2 \leq \sigma^{-1}. \quad (1)$$

Fix  $q = 1/2$ . The head  $H_\sigma = \{\mu_j : |\mu_j| > q\}$  is finite. On the Hilbert space  $\ell^2(\{j : |\mu_j| \leq q\})$ , define

$$C_\sigma = \text{diag}(\mu_j : |\mu_j| \leq q).$$

The canonical-product factorization from RH-234 shows that  $C_\sigma$  realizes exactly the complementary projection-free  $\det_2$  factor.

## 2 Variable-block theorem

**Theorem 2.1** (instantiated RH-279 certificate). *For every integer  $m \geq 2$ ,*

$$\|C_\sigma^m\|_1 \leq M_\sigma q^{m-2}, \quad \|C_\sigma^m\| \leq q^m, \quad (2)$$

$$\|C_\sigma^r\| \leq q^r \quad (0 \leq r < m), \quad |\text{Tr } C_\sigma^n| \leq M_\sigma q^{n-2} \quad (n \geq 2). \quad (3)$$

Let  $R = 7/5$  and  $m_\sigma = \lceil 4\log(1/\sigma) \rceil$ . Then

$$\limsup_{\sigma \downarrow 0} \|C_\sigma^{m_\sigma}\|_1^{1/m_\sigma} R \max\{1, qR\} \leq \frac{7}{10} e^{1/4} < 1.$$

Consequently

$$\sum_{n \geq m_\sigma} \frac{|\text{Tr } C_\sigma^n| R^n}{n} \longrightarrow 0. \quad (4)$$

*Proof.* The operator is normal and diagonal. Since  $|\mu_j| \leq q$  on the tail,

$$\sum_j |\mu_j|^m \leq q^{m-2} \sum_j |\mu_j|^2,$$

which gives every displayed norm and trace estimate. By (1),

$$\limsup (M_\sigma q^{-2})^{1/m_\sigma} \leq e^{1/4}.$$

Because  $qR = 7/10$ , the root-rate estimate follows. Finally,

$$\sum_{n \geq m} \frac{|\operatorname{Tr} C_\sigma^n| R^n}{n} \leq \frac{M_\sigma q^{-2} (qR)^m}{m(1 - qR)}.$$

For the chosen clock the right side is  $O(\sigma^{4 \log(10/7) - 1} / \log(1/\sigma))$ , whose exponent is  $0.426699 \dots > 0$ .  $\square$

### 3 Head rank and exact scope

**Proposition 3.1.** *The modulus-complete head obeys  $\#H_\sigma \leq M_\sigma/q^2 \leq 4\sigma^{-1}$ .*

*Proof.* Every head eigenvalue contributes more than  $q^2$  to the squared spectral mass in (1).  $\square$

*Remark 3.2.*  $C_\sigma$  is a normal spectral realization of the canonical-product tail. It need not be similar to a bounded physical compression, and no estimate on the ill-conditioned Riesz projectors of RH-232 follows. The theorem supplies the variable-rank tail obligation, not the head-to-monodromy coefficient bridge. Gates A–E remain false/open; no Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, or RH conclusion is asserted.